

Received October 14, 2021, accepted November 25, 2021, date of publication December 24, 2021, date of current version January 5, 2022.

Digital Object Identifier 10.1109/ACCESS.2021.3138407

Deep Learning-Based Workers Safety Helmet Wearing Detection on Construction Sites Using Multi-Scale Features

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This work was supported by Central South University.

ABSTRACT Wearing safety helmets can effectively protect workers safety on construction sites. However, workers often take off the helmets because of weak security-conscious and discomfort, then hidden dangers will be brought by this behaviour. Workers without safety helmets will suffer more injuries in accidents such as falling human body and vertical falling matter. Hence, detecting safety helmet wearing is a vital step of construction sites safety management and a safety helmet detector with high speed and accuracy is urgently needed. However, traditional manual monitor is labour intensive and methods of installing sensors on safety helmet are difficult to popularize. Therefore, this paper proposes a deep learning-based method to detect safety helmet wearing at a satisfactory accuracy with high detection speed. Our method chooses YOLO v5 as the baseline, then the fourth detection scale is added to predict more bounding boxes for small objects and the attention mechanism is adopted in the backbone of the network to construct more informative features for following concatenation operations. In order to overcome the defects caused by insufficient data, targeted data augmentation and transfer learning are used. Improvements caused by every modification are discussed in this paper. Finally, our model achieves 92.2% mean average precision, up 6.3% compared to the original algorithm, and it only takes 3.0 ms to detect an image at 640×640 . These results demonstrate the robustness and feasibility of our model. Meanwhile, the size of our trained model is only 16.3 m, which means the model is easy to be deployed. At last, after obtaining a satisfactory model, a graphical user interface (GUI) is designed to make our algorithm more user-friendly.

INDEX TERMS Safety helmet wearing detection, YOLO v5, four detection scales, attention mechanism, GUI design.

I. INTRODUCTION

With the development of urbanization, a large number of infrastructures need to be constructed, so construction sites safety raises more public attention than ever before. Many accidents can be prevented by the use of personal protective equipment (PPE) [1]. Safety helmet is one of the effective PPEs to protect workers from the hit of falling objects [2], and it is a statutory requirement all over the world to wear a safety helmet on construction sites [3]. However, safety helmet wearing tend to be ignored because of discomfort and weak safety awareness. So inspecting whether workers wearing safety helmet properly is vital for their safety and can improve safety management level. Traditional inspection

of helmet wearing on construction sites mainly includes the inspection of monitoring image and manned patrol [4]. The latter takes lots of time and labor, and manual monitor requires inspectors to stare at the screen for a long time which may cause misjudgments because of tiredness. Driven by this, new technologies for detecting workers safety helmet wearing condition on construction sites are developing rapidly with the assistance of sensors and image analysis methods.

Various sensors were utilized for safety helmet wearing detection, such as chinstrap sensors, three-axis accelerometer sensor [5], radio frequency identification (RFID) [6] and pressure sensor [7]. However, these methods increase the detection investment and can be regarded as intrusion to workers. So, workers tend to be reluctant to wear safety helmet with above sensors because of privacy and health issues [8]. As a result, non-intrusive methods are

The associate editor coordinating the review of this manuscript and approving it for publication was Oğuzhan Urhan[✉].

more acceptable, and among them, computer vision holds immense potential to detect safety helmet wearing condition [9]. Many computer vision-based algorithms have been proposed to detect objects, such as Gaussian mixture model (GMM) [10], histogram of oriented gradient (HOG) [11], and support vector machine [12]. These algorithms were also used in safety helmet detection to deal with the high fatality rates confronting the construction industry [13]. Liu *et al.* [14] used skin color and hu moments to recognize helmet. Rubaiyat *et al.* [15] used HOG to detect workers, then color-based method and circle hough transform were applied to complete safety helmet detection. Wu *et al.* [16] extracted features of safety helmets with different colors and then used a hierarchical support vector machine to classify workers safety helmet wearing condition. However, above traditional detection methods have their drawbacks. GMM can not distinguish the category of the foreground well, and hand-designed features such as color and HOG have difficulties to effectively capture high and mid-level representations. So detectors based on these methods have poor generalization ability and they are hard to be applied in practical engineering.

Thanks to the development of computer technology, applying GPU in large-scale parallel computing makes training large deep neural networks possible [17], [18]. In the field of object detection, a series of methods based on deep learning were developed, and convolution neural networks (CNNs) are most commonly used because of their superiority in high-level feature extraction. As a result, they gradually replace the traditional detection methods in image analysis [19]. Generally, CNN-based object detection methods can be divided into two categories. One is two-stage detectors, they firstly extract a series of candidate regions where there might be objects, then CNN detectors are applied to objects classification and location. Their representative networks are region-based convolution neural networks (R-CNN) [20] and its improved networks: Fast R-CNN [21] and Faster R-CNN [22]. The other is single-stage detectors, they directly predict class probabilities and bounding box coordinates according to the CNN features by treating object detection as a regression problem. Their representative networks are single shot multibox detector (SSD) [23], you only look once (YOLO) [24] and its improved networks [25]–[27]. Above object detectors continue to achieve better results in major datasets.

The development of CNN-based detectors has motivated deep learning-based safety helmet wearing detection methods [28], [29], and many researchers considered deep learning-based methods as a vital measure to deal with the construction safety management issue [13]. Two-stage detectors were first used in detecting helmet wearing. Fang *et al.* [9] developed an automatic non-hardhat-use detector based on Faster R-CNN and achieved an accuracy rate between 90.1% and 98.4% in different scenarios. However, it needs around 0.2s to detect an image, which is unable to meet the real-time requirement. Gu *et al.* [30]

used multi-scale training, increasing anchors strategies and online hard example mining to improve Faster R-CNN. The safety helmet detection accuracy finally is increased by 7% comparing to the original algorithm, but the operating speed was not given. Recently, more and more researchers chose single-stage detectors to complete safety helmet detection task. Shi *et al.* [31] used image pyramid structure to extract multi-scale feature map and combined it with YOLO v3. Wu *et al.* [32] adopted densely connected convolutional networks [33] to instead the original backbone of YOLO v3 and got better detection results under the same detection time. Shen *et al.* [8] detected safety helmet based on a single-stage face detector [34], then acquired a face-to-helmet regression model. Li *et al.* [2] chose SSD algorithm to meet the real-time requirement and added MobileNet [35] to reduce computation load. Wang *et al.* [4] proposed a new objective function to improve YOLO v3 and applied it in helmet detection. As single-stage detectors, they usually have lower accuracy rates and faster operating speed compared with two-stage detectors [36].

All above researches show that it is meaningful to develop deep learning-based methods to detect safety helmet wearing, which can help reduce human and material resources, prevent missed and false detection caused by human reasons, and lay a foundation for the next step. However, both two-stage and single-stage detectors have their limitations, they do not have enough speed or accuracy. Fortunately, many features have been proposed recently to improve CNN accuracy, some among them have universality, such as weighted residual connections (WRC) [37], cross stage partial connections (CSP) [38], self adversarial training (SAT) and so on. They can help single-stage detectors achieve state-of-the-art results [27]. Considering the trade-off between the speed and accuracy, this paper proposes a method to detect safety helmet wearing condition based on YOLO v5 [39], which can compute fast enough with a satisfactory accuracy.

Currently, the deep learning-based methods are data-driven, which means the performance of object detection algorithms depends on the datasets used for training, but the number of safety helmet images is limited. In order to improve the performance of YOLO v5 on the task of safety helmet wearing detection, in this research, targeted data augmentation and transfer learning were used, at last, a GUI is designed for our model.

The main purpose of this paper is to propose a fast, high-precision and easy-to-use network to realize the automatic detection of helmet wearing status in the construction site. At the same time, the model should have strong generalization ability, not limited to helmet detection in a specific scene, and can well deal with small targets and multi-scale challenges. Above factors are taken into fully consideration in the dataset used in this paper, and various scenes including helmet images of various challenges are collected.

The rest of this paper is organised as follow: Section II presents the principle of the series of YOLO detectors and our modifications on YOLO v5 and datasets. Section III

shows the implementation of safety helmet detection GUI. Section IV describes the details of the experiments designed in this paper, which include the computer configuration, datasets and evaluation metrics. Then the experimental results and discussion are presented in Section V. Finally, Section VI draws conclusions and gives prospects of future research.

II. METHODOLOGY

This section first simply introduces YOLO detectors, then the improvements are proposed according to the object function of bounding box regression. Meanwhile, in order to overcome the defect of insufficient data, targeted operations are conducted. The overview of the safety helmet detection method is shown in Fig. 1.

A. THE STRUCTURES OF YOLO DETECTORS

The series of YOLO detectors frame detection as a regression problem. Input images are divided into $S \times S$ grids as Fig. 2, the image is divided into 7×7 by the red lines.

If the center of an object falls into a grid cell, that grid cell is responsible for detecting that object. In Fig. 2, the center of the object falls in row two column five, hence this grid cell takes the responsibility for detecting helmet wearing. S is defined as 7 in YOLO v1 [24], and each grid cell predicts 2 bounding boxes, so only 98 boxes can be predicted in all. In Fig. 2, the yellow bounding box is one of the two predictions of the grid cell, and the green one is the ground truth (GT). In order to enable the model to predict more boxes and improve the performance of algorithm, YOLO v2 [25] draws the idea of anchor boxes from Faster R-CNN. First, input images of 416 are downsampled by the convolutional layers with a factor of 32, so YOLO v2 predicts boxes on a feature map of 13×13 . Then, for each cell, it predicts 5 bounding boxes, which is decided according to anchor boxes clustered from training set.

However, above YOLOs struggle with small objects. To solve this problem, from the third version of YOLO, this series of detectors predict objects at 3 different scales. YOLO v3 [26] first takes the feature maps from two previous layers of the original prediction layer and upsample it by two times, then a feature map from earlier layer is taken to be concatenated with the upsampled features. A few convolutional layers are added to process the combined feature map to predict in the second scale. These steps are performed once more to predict in the third scale. The final structure of YOLO v3 is shown in Fig. 3.

YOLO v4 [27] and v5 [39] adopt similar structures with YOLO v3 and modify the state-of-the-art methods such as cross-iteration batch normalization (CBN) [40], path aggregation network (PAN) [41], spatial attention module (SAM) [42] and so on, which make the network become efficient and suitable for single GPU training. Compared to YOLO v4, YOLO v5 is more flexible and smaller, so it is chosen to detect the safety helmet wearing to integrate with the hardware in the future.

B. INCREASE OF REASONABLE BOUNDING BOXES FOR SMALL OBJECTS

Object detection algorithms such as SSD and the series of YOLO train model on big datasets like MS COCO [43] which has 80 object categories. Generally, more categories mean more pressure to GPU during training and this limits the improvements of the network structure because all experiments in this paper are conducted on single GPU. Fortunately, in some specific detection tasks, we do not need to identify so many categories. In this paper, only two object categories need to be detected, so more complicated structure can be used to improve the performance of the network.

YOLO v5 model has four sizes according to the depth and width of the structure, they are s, m, l and x, and their complexity increases in turn. On our dataset, only size s and m can be trained with single GPU, and almost all GPU memory is occupied while training size m of YOLO v5 model, this paper hence chooses size s of YOLO v5 model as baseline to detect safety helmet wearing. Compared to other YOLO application [32], the safety helmet wearing detection in our project often requires the model to recognize an object which occupies only a small proportion in an image. Meanwhile, the detection meets challenges such as scale variances. Due to the limitation of the depth and width, YOLO v5 s does not perform well in solving above problems. Therefore, we decide to improve the structure so that the model can detect small targets and face the challenges of scale change better. And we found inspiration from the loss function, which is an important part of the algorithm.

The object function of bounding box regression in YOLO v5 changes from sum of squared error (SSE) loss to complete intersection over union (CIoU) loss [44], the schematic illustration of CIoU is shown in Fig. 4, where the green bounding box is the ground truth B^{gt} and the yellow one is a predicted bounding box B and the CIoU loss is calculated as follow:

$$L_{CIoU} = 1 - IoU + \frac{d^2}{l^2} + \alpha v \quad (1)$$

where IoU is the intersection over union and is defined in Equation 2.

$$IoU = \frac{|B \cap B^{gt}|}{|B \cup B^{gt}|} \quad (2)$$

In Equation 1, d is the distance between the two central points c^{gt} and c of the two boxes, and l is the diagonal Length of the smallest enclosing box covering two boxes. v indicates the aspect ratio consistency, and it is calculated according to the width w^{gt} , w and height h^{gt} , h of the ground truth and bounding box, its definition is shown in Equation 3. Then, the weight parameter α is defined in Equation 4.

$$v = \frac{4}{\pi^2} (\arctan \frac{w^{gt}}{h^{gt}} - \arctan \frac{w}{h})^2 \quad (3)$$

$$\alpha = \frac{v}{(1 - IoU) + v} \quad (4)$$

These formula shows that the result of object function is closely related to predicted bounding box, so predicting

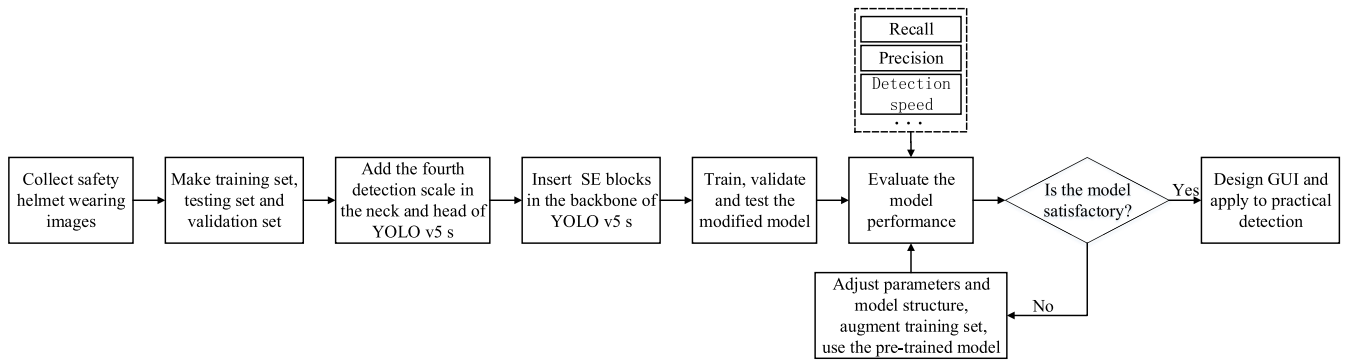


FIGURE 1. Flowchart for safety helmet detection.

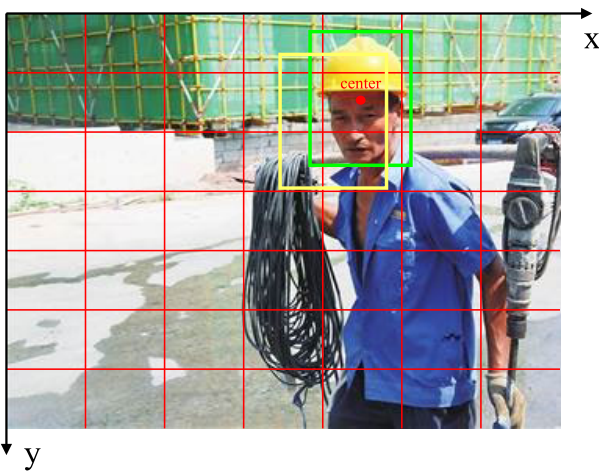


FIGURE 2. Schematic diagram of image division.

TABLE 1. The details of four scales.

Scales	Size	Anchors
first scale	13×13	(11, 13) (17, 19) (23, 26)
second scale	26×26	(30, 34) (38, 43) (47, 53)
third scale	52×52	(58, 67) (76, 88) (104, 121)
fourth scale	104×104	(149, 173) (229, 265) (414, 478)

more reasonable boxes can help train the network better. Hence we make improvements on the neck and prediction head of YOLO v5 s as shown in Fig. 5 and our modifications are labelled by the red dotted lines. Compared to the original structure, the concatenation operation mentioned in Section II-A is conducted once more to predict objects in the fourth scale, which means the feature maps are upsampled again, so the fourth scale has smaller receptive field than the first three scales. This operation can enhance the model's performance on predicting bounding boxes for small objects. Details of four scales to predict objects are listed in Table 1.

C. ATTENTION MECHANISM FOR STRENGTHENING THE REPRESENTATIONAL POWER OF THE NETWORK

As shown in Fig. 5, there are many concatenation operations extracting feature maps from previous layers, and we

hope that the useful informative features can be exploited as much as possible, attention mechanism is hence applied in the structure of the network to make the feature maps have more representational information before concatenation operations. After a series of experiments, two squeeze-and-excitation (SE) blocks [45] are hence inserted into the backbone of the structure to bring more improvements at negligible additional computational cost. In order to avoid the influence of subsequent convolution operation, SE blocks are just inserted in front of the concatenation operations as shown by the red dotted in Fig. 5.

D. TARGETED OPERATIONS REGARDING DATASET

The dataset used in this paper has two object categories, they are person and hat. The details of the dataset will be presented in Section IV. At first, to avoid class imbalance problem [46], we trained the model on the training set, in which the numbers of input bounding boxes pertaining different classes are approximately equal. Then according to the model performance on testing set (The details will be mentioned in Section V), we increased the number of objects belongs to the class which the model had lower average precision (AP) for in training set. Meanwhile, to further improve the detection accuracy under limited training samples, a pre-trained model is used and transfer learning approach is adopted.

III. SAFETY HELMET DETECTION GUI DESIGN

After obtaining satisfactory model, a GUI is designed to make the safety helmet wearing detection more user-friendly. The interface of the GUI is shown in Fig. 6, and users can simply detect safety helmet wearing by select files and clicking the detect button. The whole process of the GUI is shown in Fig. 7. Six major steps are summarized as follows:

- 1) *Open GUI*: The interface shown in Fig. 6 will be instantly displayed on the screen once the GUI is opened.
- 2) *Select Files*: Users can select files by clicking the Browse button to choose the folder where the files are located, then image and video files under the folder will be listed in the list box for users to select. Users can

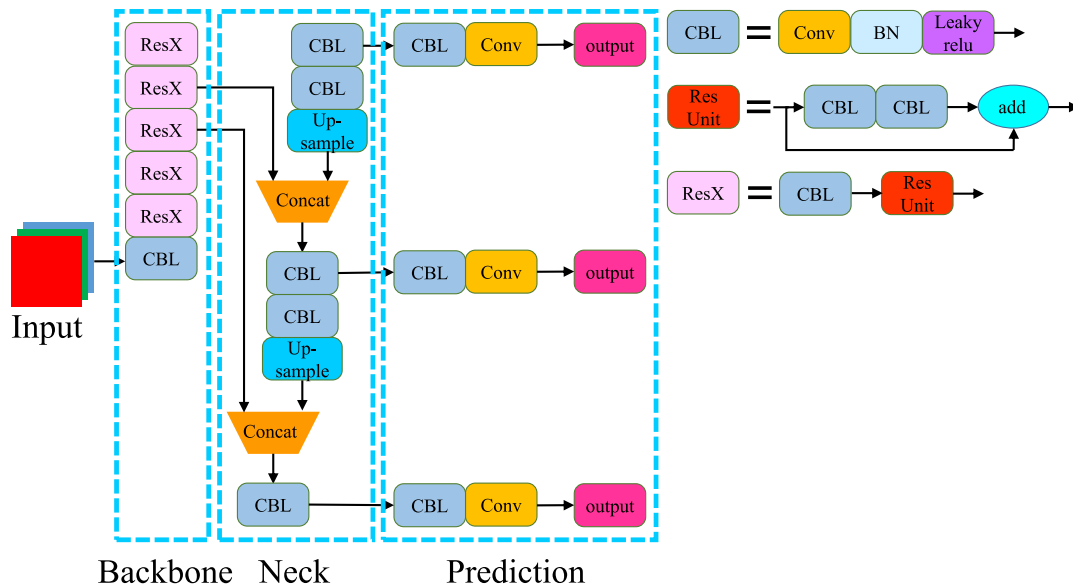


FIGURE 3. YOLO v3 structure.

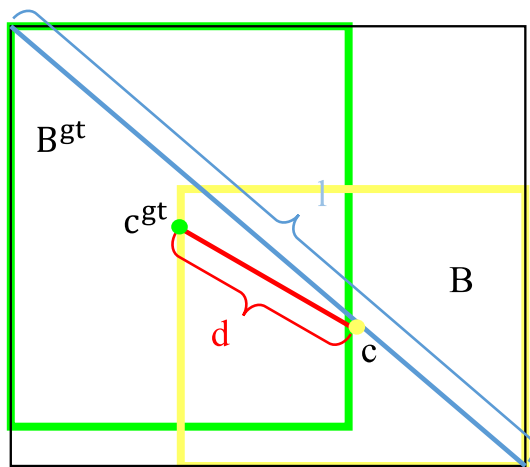


FIGURE 4. CIoU loss for bounding box regression.

also obtain image data from the camera by clicking the Obtain from camera button. A thumbnail will be shown on the right side of the list box once the file is selected.

- 3) *Detect Safety Helmet Wearing*: Users can detect safety helmet wearing on selected files by clicking the Detect button. Then our model will process the images with default settings and the detection results will be displayed directly on the right side of the GUI. If the file is a video, every frame will be detected and played on the right side. Then users can drag the slider at the bottom to control the display process of the detected video. The Pause button can be used to pause the video, and the video can be resumed by clicking the button again.
- 4) *Adjust Confidence Threshold*: Users can set object confidence threshold by dragging the slider or entering the threshold into the input box directly, and the detection

results will change with adjustment instantly. This step can help filter out some false detection results. Users can skip this step and the default confidence threshold is 0.4.

- 5) *Adjust IOU Threshold*: IOU threshold needs to be raised when there are more than one bounding box predicted on one object. And it needs to be decreased when the overlapping of multiple objects leads to missed detection. The adjustment can also make influence on detection results immediately. This step can be skipped as well and the default confidence threshold is 0.5.
- 6) *Save*: Users can select a path to save the detection results by clicking the Save button.

IV. DATA SCOPE AND EXPERIMENTAL DESIGN

A. COMPUTER CONFIGURATION

All experiments in this paper were conducted on a workstation with an NVIDIA RTX 2080Ti GPU. CUDA, cuDNN, OpenCv, Pytorch were used to realize the detection model. The details of computer configuration are shown in Table 2.

B. DATASETS

Since there are not many mature deep learning-based methods used in safety helmet wearing detection, and the datasets they used usually have some specific features which are not suitable for other safety helmet wearing detection tasks, we objectively collect the helmet and person images which include various scenarios from the internet and construction sites. In order to improve the generalization ability of the model, challenges of object detection are also considered, and some of them are shown in Fig. 8. We randomly separate 80% all collected images as a training set and 80% rest images as a testing set, then remained images belong to validation set.

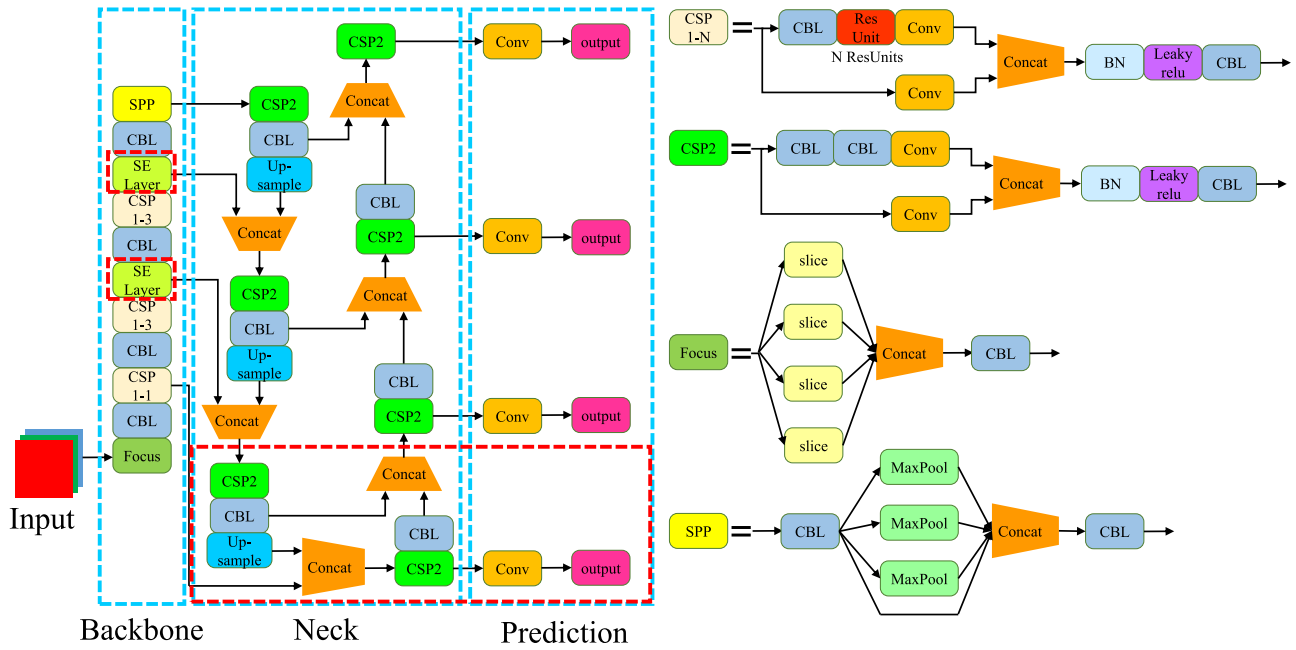


FIGURE 5. Modified YOLO v5 s structure.

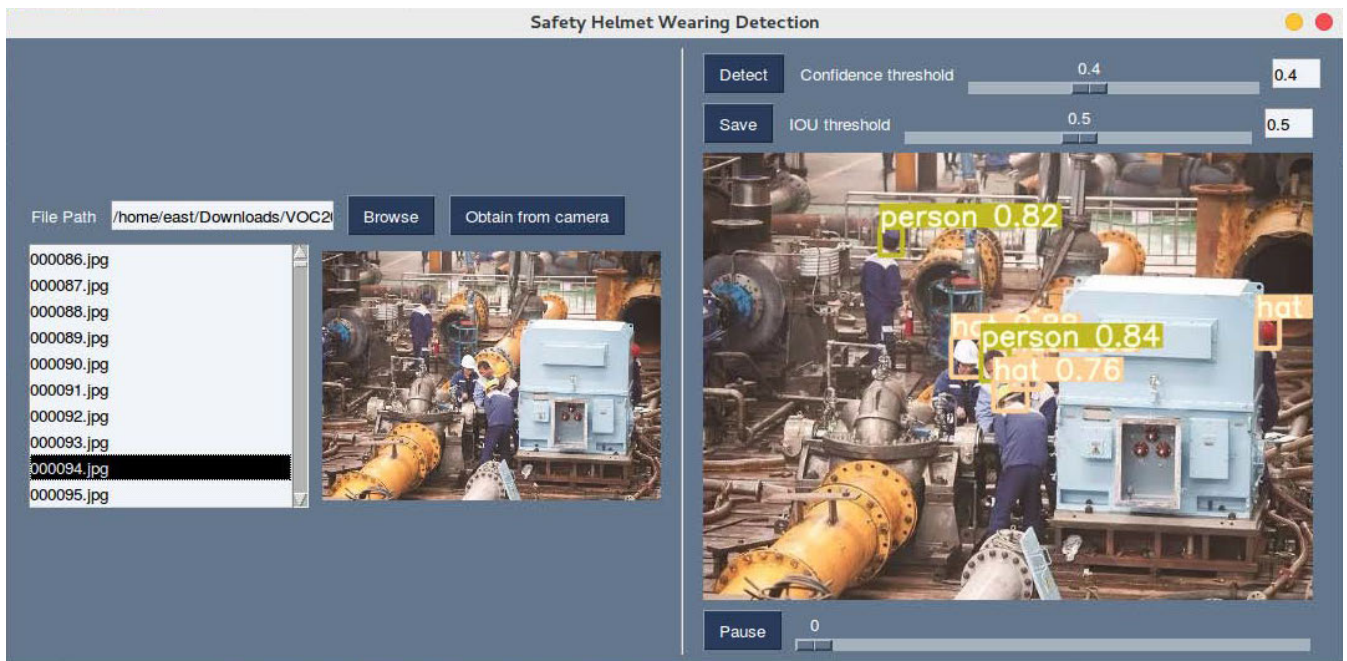


FIGURE 6. Interface of safety helmet detection GUI.

The exact numbers of samples used for training, validation and testing are shown in Table 3. In order to further improve model's performance, more samples are needed. According to the model's results on testing set (mentioned in Section V), more person images are added into the training and validation set, the testing set remains unchanged for comparison. The exact numbers of samples used for training, validation in modified dataset are shown in Table 4.

C. EVALUATION METRICS

Each hat and person in the datasets is labelled manually, and the produced bounding boxes are defined as the ground-truth, then a predicted bounding box can be regarded as correct when the IoU value of the predicted box and corresponding labelled box exceeds the threshold, in this paper, the threshold is defined as 0.5. Hence the model's performance can be evaluated by calculating the following metrics:

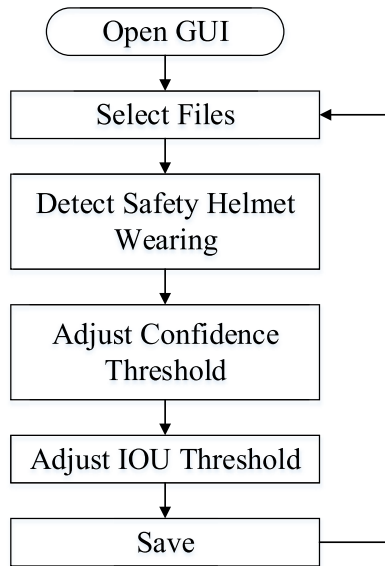


FIGURE 7. Flowchart of safety helmet detection GUI.

TABLE 2. Computer environment for the experiments.

Configuration	Type
System	Ubuntu 18.04.5 LTS
CPU	Intel Xeon(R) Gold 5218: 2.30 GHz, 16 cores, 32 threads
Memory	62.6 GB
GPU	NVIDIA GeForce RTX 2080 Ti: 11264 MB, 14000 MHz
Disk	Jesis X760S 512 GB + TOSHIBA MG04ACA4 4.4 TB
OS type	64-bit
Python	version 3.6.9
Nvidia driver	version 450.36.06
CUDA	version 11.0
cuDNN	version 8.0.2
Pytorch	version 1.5.1
OpenCV	version 4.2.0

TABLE 3. The number of objects of different classes in the initial dataset.

Categories	Training set	Validation set	Testing set
hat	7137	371	1539
person	7141	525	1416

TABLE 4. The number of objects of different classes in the modified dataset.

Categories	Training set	Validation set
hat	7137	371
person	35531	2003

- 1) The precision-recall (PR) curve, which is drawn according to the precision (y-axis) and recall (x-axis) for different probability thresholds. Compared with single variable, it can provide a more comprehensive understanding of the relationship between the precision and recall for comparison of different model. The precision and recall are defined in Equation 5 and



FIGURE 8. Some challenges in object detection.

Equation 6:

$$precision = \frac{TP}{TP + FP} \quad (5)$$

$$recall = \frac{TP}{TP + FN} \quad (6)$$

For each class, the true positive (TP) is the number of the objects that are correctly predicted by the model; the false positive (FP) is defined as the number of predicted objects that are actually other contents; the false negative (FN) denotes the number of the ground truth that are not correctly predicted by the model. In general, the closer the curve to the upper right corner, the better the model's performance.

- 2) The mean average precision (MAP), which is the mean value of the average precision (AP) of all classes, and it is defined in Equation 7:

$$MAP = \frac{\sum_{i=1}^n AP_i}{n} \quad (7)$$

where n is the number of the classes, and for each class, average precision is the area under the corresponding PR curve. Normally, the higher the MAP, the better the model's performance.

- 3) The mean detection time (MDT), which is the mean value of the time used to process an image. In this paper, each image in testing set is reshaped to 640×640 when processed by the model and there are 32 test images in a batch. Generally, the smaller the MDT, the faster the model's speed.

V. EXPERIMENTAL RESULTS AND DISCUSSION

In this paper, the main model improvements include the fourth scale (FS), 2 SE blocks (SB), training and validation datasets augmentation (DA) and pre-trained model (PT). The influence of them on models are evaluated in terms of performance and computing speed on the same testing set by calculating the three metrics mentioned in Section IV-C. And we also provide the comparison between our method and other representative state-of-the-art (SOTA) model in the field of object detection at last.

A. EVALUATION OF THE MODIFIED YOLO v5 MODEL'S PERFORMANCE

We first evaluate the performance improvement caused by the modifications on the model. The loss curves while training the models and the PR curves while testing the models are shown in Fig. 9.

It shows that the modifications of the model can make it converge and perform better on loss curves and PR curves respectively. Besides, the addition of the 2 SE blocks can provide better performance on PR curves with little impact during training process. For each class, the corresponding AP are listed in Table 5 and it shows that the class person has lower AP, then we add more images belongs to the class person into training and validation datasets as mentioned in Section II-D, the results show that the AP of class person increases by 5.1% and the AP of class hat also raises slightly, which means it is useful to conduct targeted data augmentation on the category which the model has lower accuracy on. The pre-trained model is also used to further improve the performance of the model, and the comparison is shown in Fig. 10.

Although the loss converges at a higher value due to the increased images, on PR curves (Fig. 10b), the models after targeted operations mentioned in Section II-D perform better on the same testing set, and they converge faster on loss curves (Fig. 10a). Meanwhile, Targeted operations are applied on the size s and m of YOLO v5 to further prove our modifications on YOLO v5 s are effective, and the results are shown in Fig. 11.

From Fig. 11, we can find that under the same training strategies, the model modified by us performance better than the original size s and m of YOLO v5 both on the loss curves and PR curves. The precision of PR curves in above figures is the mean value of the precision value of class hat and person, and separate PR curves for each class are shown in Fig. 12, the AP for each class is listed in Table 5. Compared with the original YOLO v5 s, our model increases by 6.3%.

For class hat, YOLO v5 m with DA and PT has higher AP than other models in this paper. However, under the same training strategies, the models modified by us is much better on the AP of class person, meanwhile, it has the highest MAP on testing set among the models in this paper. What's more, in practical application, it is more necessary to detect people who do not wear safety helmet since most people wear safety

TABLE 5. Models comparison regarding AP.

Model	AP		MAP
	hat	person	
YOLO v5 s	0.898	0.821	0.859
YOLO v5 s with FS	0.904	0.839	0.871
YOLO v5 s with FS and SB	0.912	0.839	0.876
YOLO v5 s with FS, SB and DA	0.914	0.89	0.902
YOLO v4 with DA and PT	0.892	0.866	0.879
YOLO v5 s with DA and PT	0.919	0.88	0.9
YOLO v5 m with DA and PT	0.927	0.891	0.909
YOLO v5 s with FS, SB, DA and PT	0.926	0.917	0.922

helmets on construction sites. On this condition, higher AP for class person is more meaningful, and some test results on real scenes are shown in Fig. 13. It can be seen that for some small objects original YOLO v5 s will make missed and false detection, however, our model can effectively solve these problems.

B. EVALUATION OF THE MODIFIED YOLO v5 MODEL'S COMPUTING SPEED

The MDT of each model is listed in Table 6 and it contains three aspects: time used for inference, time used for non maximum suppression (NMS) and total time. Where inference is the process of generating all bounding boxes, and NMS is the process of removing the invalid and redundant bounding boxes. All time used for inference and NMS is the total time. It can be seen that the main difference of different models' MDT is about the inference time, and their NMS times are approximately the same. Compared with the original YOLO v5 s, our model has longer inference time, but it is faster than YOLO v5 m. Besides, real-time detection requires the speed to exceed 30 frames per second (FPS) and each model in Table 6 is far beyond this standard. Hence, we think it's worth sacrificing some speed to reduce the missed and fault detection and get performance improvements.

C. COMPARISON STUDIES

In this paper, we select some representative SOTA detection models from different categories to conduct comparison experiments. For anchor-based models, which need to explicitly define priori anchors, YOLO v4 [27] is chosen as the representative, it is also a sing-stage detector, and achieves high accuracy and fast speed on MS COCO dataset [43]. As for anchor-free models, which do not need to explicitly define priori anchors, we choose Centernet2 [47] as the representative, the design of it takes into account the advantages of both two-stage and single-stage methods, and it also show well performance on MS COCO dataset. Excepted for above CNN-based detectors, many researchers have studied transformer-based models recently, they also show strong performance. In this category, we chose Cascade Mask R-CNN with Swin transformer as the backbone (Swin-CMR) [48]. Finally, the small size of YOLO v4, Centernet2 and Swin-CMR are trained and tested on the

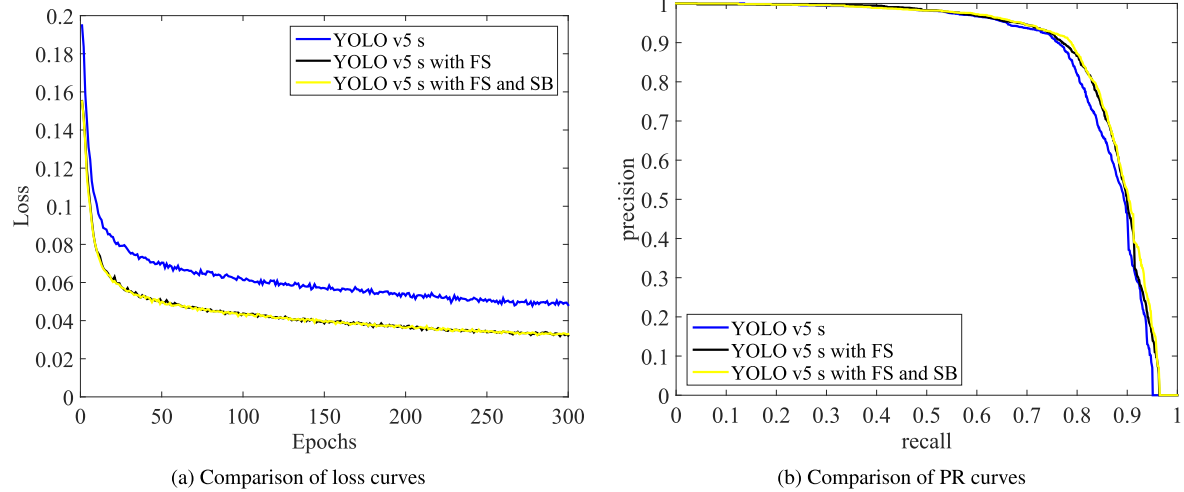


FIGURE 9. Performance of models with different modifications.

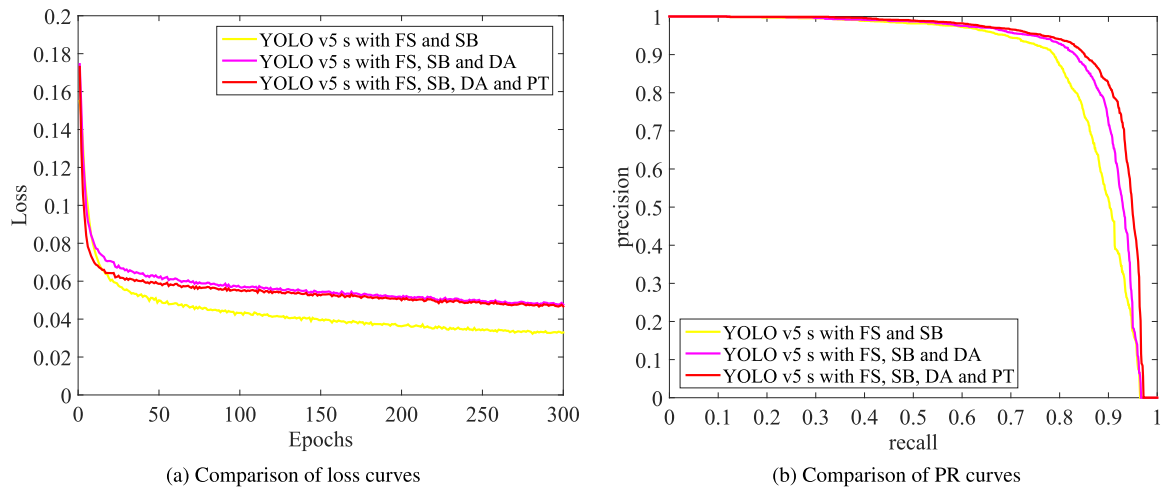


FIGURE 10. Performance of models with different training strategies.

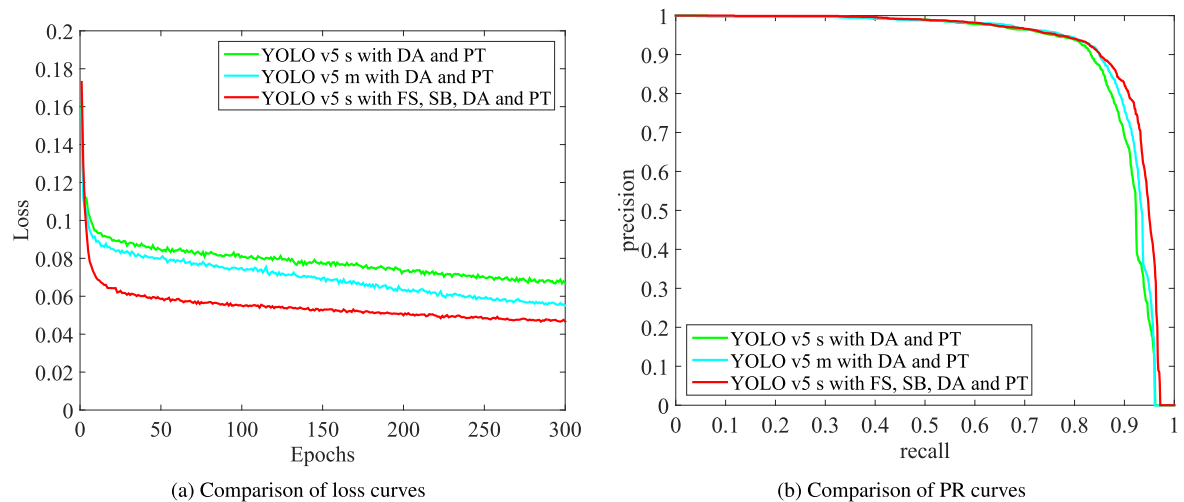


FIGURE 11. Performance of different models under the same training strategies.

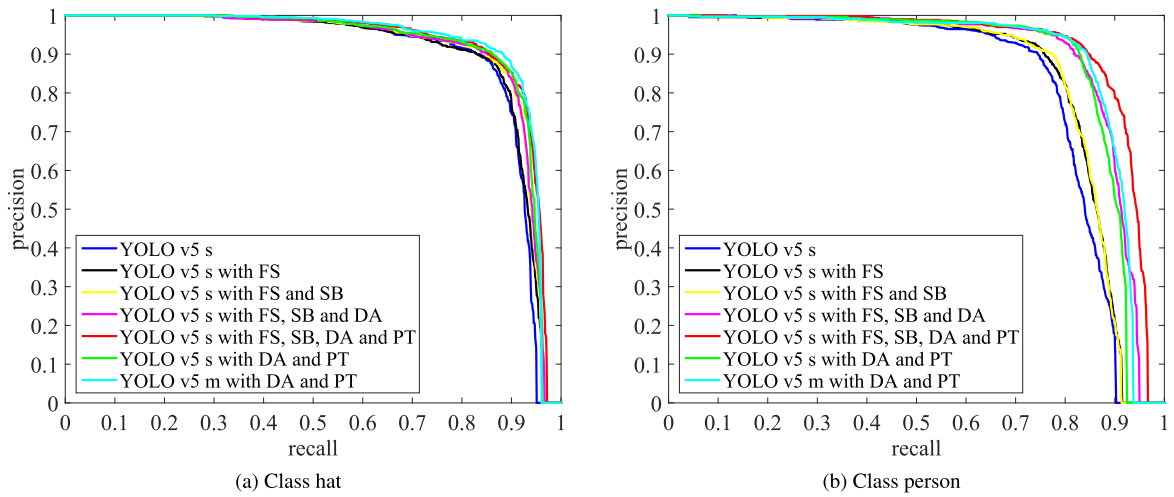


FIGURE 12. Performance comparison of models on class hat and person.



FIGURE 13. Models comparison in real scenes. (a), (c) and (e) are tested by YOLO v5 s, and (b), (d) and (f) are tested by YOLO v5 s with FS, SB, DA and PT.

TABLE 6. Models comparison regarding MDT.

Model	MDT(ms)			FPS
	inference	NMS	total	
YOLO v5 s	1.4	0.9	2.3	435
YOLO v5 s with FS	2.1	1.0	3.1	323
YOLO v5 s with FS and SB	2.1	1.0	3.1	323
YOLO v5 s with FS, SB and DA	2.1	1.0	3.1	323
YOLO v5 s with FS, SB, DA and PT	2.1	0.9	3.0	333
YOLO v4 with DA and PT	6.5	1.0	7.5	133
YOLO v5 s with DA and PT	1.4	0.9	2.3	435
YOLO v5 m with DA and PT	2.7	0.8	3.5	286

same safety helmet dataset with our method to conduct the comparison experiments. And each model is trained until

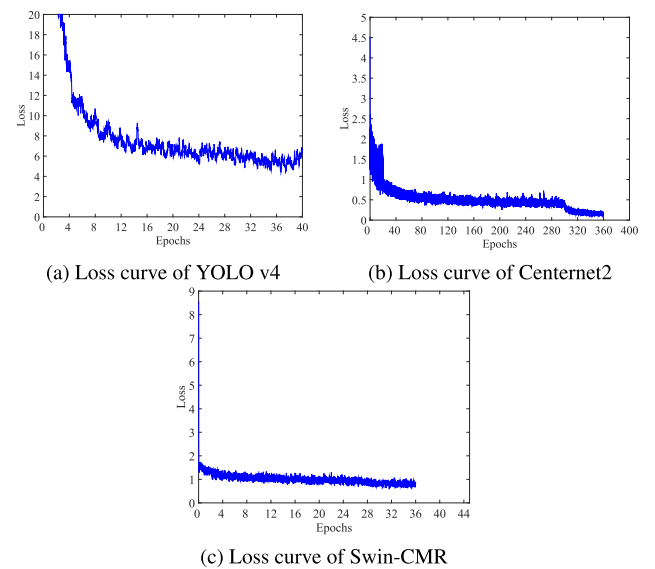


FIGURE 14. Loss curves of YOLO v4, Centernet2 and Swin-CMR.

TABLE 7. MAP and speed comparison.

Model	AP hat	person	MAP	MDT(ms)	FPS	Model size(MB)
YOLO v4	0.898	0.821	0.859	7.5	133	256
Centernet2	0.917	0.90	0.909	27.1	37	379.3
Swin-CMR	0.928	0.914	0.921	49.8	20	1331
Our model	0.914	0.89	0.902	3.1	323	16.3
Our model with PT	0.926	0.917	0.922	3.0	333	16.3

the loss curves does not fail, the loss curves are shown in Fig. 14. Transfer learning is not used in the comparative experiments because the pre-trained model of each algorithm is obtained on different dataset sets, and some models, such as Swin-CMR, can not use the pre-trained model easily due to the different number of classes.

We compare above models in terms of accuracy and speed as the same, and the results are shown in Table 7. It can be seen that our model has the fastest speed and the smallest size compared with other models. The MAP is not the highest, but it is also satisfactory, and it can be higher with the pre-trained model (Section V-A). All in all, our model achieves a better trade-off between detection accuracy and speed, and it can be applied to practical projects more easily.

VI. CONCLUSION

In this paper, we presented a fast, accurate and easy-to-use method to detect safety helmet wearing condition. And the main contributions are as follows: (a) We design the fourth detection scale based on YOLO v5 s. This modification can help reduce the missed and false detection caused by the challenges of small objects and scale variances. (b) Two SE blocks was added in the backbone of the modified model to further improve accuracy at negligible additional computational cost. (c) Targeted data augmentation and pre-trained model were adopted to overcome the defects caused by insufficient data. (d) A GUI is designed to make our method more user-friendly after obtaining the satisfactory model, which means our model can be applied to practical engineering easily. Our model were tested on the same testing set which contains various challenges such as scale variances, small objects and illumination effect after every modification. Finally, our modified YOLO v5 s reaches to 92.2% on MAP, up 6.3% compared to the original YOLO v5 s. Meanwhile, its detection speed meet and exceed the real-time requirement, it only takes 3.0 ms on average to detect an image at 640×640 . The evaluation of performance and speed proves the rapidity and reliability of our model. Fast speed ensure that workers can be reminded to wear safety helmets in time, while high accuracy can reduce missed and false detection, both of them are essential in improving workers' safety situation. Besides, the targeted data augmentation used in this paper also has reference value when the data used for training is not sufficient.

In practical application, the images, videos and monitoring screen can be shown on our GUI. Users can easily adjust the threshold according to actual needs to obtain better safety helmet wearing detection results. When there are workers without safety helmets, the monitoring personnel will be reminded to take the next step. Compared with traditional methods such as manual monitoring, the method in this paper does not require the monitoring personnel to stare at the screen all the time. They only need to pay attention to the cue of the appearance of workers without safety helmets, which greatly reduces the energy consumption of the monitoring personnel and the number of monitoring personnel required. At the same time, because this method can better deal with the challenges of small and multi-scale objects, it can detect objects far away from the camera. As a result, our method increase the monitoring range of the camera, which reduces the number of patrol personnel and cameras to be arranged.

Further improvements of our modified YOLO v5 will include collecting more safety helmet wearing and person

images on different occasions to enhance the robustness and popularity of the model. More new features which are said to improve CNN performance could be tried on our modified model. Since in practice application, helmet wearing detection models are usually used to detect video, keeping the stability of the predicted bounding boxes according to the inter-frame correlation will be investigated [49]. For the lightweight and deployment of the model, pruning will be carried out. Finally, reminding workers to wear safety helmet on construction sites in time will be investigated in the future.

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